

# Recursive Genomic Q-Former: Parameter-Reduction vs. Performance Trade-off

| Task details   |                                                                                                                 |
|----------------|-----------------------------------------------------------------------------------------------------------------|
| <b>Task</b>    | cancer_type — 4-way tumor-of-origin classification from bulk gene-expression profiles (TCGA RNA-seq).           |
| <b>Dataset</b> | Unified TCGA cohort: 2,738 samples $\times$ 20,530 genes; top 4,000 highly-variable genes used as input tokens. |
| <b>Classes</b> | Breast (BRCA, 867), Head-neck (HNSC, 423), Lung (LUNG, 943), Prostate (PRAD, 505).                              |
| <b>Split</b>   | 70/15/15 train/val/test, <b>523 test samples</b> , seed 42.                                                     |
| <b>Model</b>   | Recursive Genomic Q-Former: $d_{\text{model}} = 128$ , 256 marker query slots, recursion depth 4.               |
| <b>Metric</b>  | Test accuracy and macro-averaged F1.                                                                            |

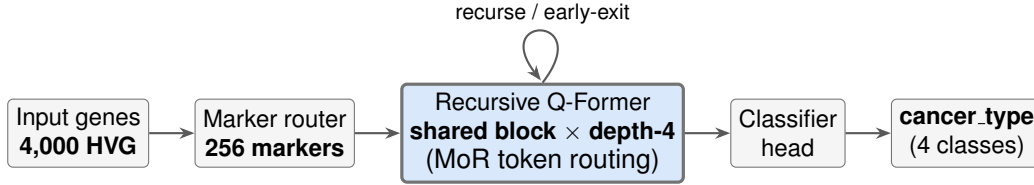


Figure 1: Pipeline. The shared transformer block (blue) is reused across recursion steps; MoR routes tokens to a per-token depth instead of a fixed pass.

All results are on the **cancer\_type** classification head (4 classes: BRCA, HNSC, LUNG, PRAD; 523 test samples), with a fixed recursion depth of 4,  $d_{\text{model}} = 128$ , 256 marker query slots, and seed 42. The “regular” recursive model shares one transformer block across all recursion steps; the comparison models either un-tie those weights (*Independent*) or add token-adaptive routing (*MoR*).

**Table 1: Parameters and performance for all three model types.** One view of the full trade-off: parameter budget (recurrent-block params, total params, and the independent/shared ratio) alongside all three performance metrics (accuracy, macro-F1, weighted-F1). Sharing the transformer block cuts recurrent params by  $\sim 4\times$  for a 0.57-point accuracy cost; Mixture-of-R recursions (MoR) keeps that small budget yet recovers the gap to 0.19 points and is best on macro-F1.

| Model                       | Weights | Routing | Parameters     |                |                                | Performance   |               |               |
|-----------------------------|---------|---------|----------------|----------------|--------------------------------|---------------|---------------|---------------|
|                             |         |         | Recurrent      | Total          | Ratio                          | Acc.          | Macro-F1      | Wtd-F1        |
| Independent stack           | untied  | none    | 530,432        | 1,109,897      | $3.99\times$                   | <b>0.9904</b> | 0.9891        | <b>0.9904</b> |
| <b>Recursive (shared)</b>   | shared  | none    | 132,992        | 712,457        | $1.00\times$                   | 0.9847        | 0.9871        | 0.9847        |
| <b>MoR (token-adaptive)</b> | shared  | token   | <b>132,992</b> | <b>712,457</b> | <b><math>1.00\times</math></b> | <b>0.9885</b> | <b>0.9892</b> | <b>0.9885</b> |

**Read-out.** The shared recursive model uses **74.9% fewer** transformer parameters than the independent stack (132,992 vs. 530,432) for a **0.57-point** accuracy cost. Adding MoR token routing

closes that gap to **0.19 points** at the same small parameter budget, and is the only configuration that also reduces FLOPs.

**Table 2: Parameter scaling vs. recursion depth.** Shared-weight recursion keeps the transformer parameter count essentially flat as depth grows, whereas an independent stack grows linearly. The reduction factor reaches nearly 8 $\times$  at depth 8.

| Recursion depth | Shared params | Independent params | Reduction factor |
|-----------------|---------------|--------------------|------------------|
| 1               | 132,608       | 132,608            | 1.00 $\times$    |
| 2               | 132,736       | 265,216            | 2.00 $\times$    |
| 4               | 132,992       | 530,432            | 3.99 $\times$    |
| 6               | 133,248       | 795,648            | 5.97 $\times$    |
| 8               | 133,504       | 1,060,864          | 7.95 $\times$    |

**Table 3: Compute trade-off (depth-4 variants).** Early-exit / adaptive recursion lowers the *effective* mean depth and FLOPs per sample relative to a fixed depth-4 pass, at equal parameter count. MoR achieves the lowest compute and the best F1.

| Configuration                  | Transf. params | Mean depth  | MFLOPs/sample | Test acc.     |
|--------------------------------|----------------|-------------|---------------|---------------|
| Fixed depth-4 (no early exit)  | 132,480        | 4.00        | 268.4         | 0.9847        |
| Recursive (shared, early-exit) | 132,992        | 2.75        | 161.5         | 0.9847        |
| No recursive refine            | 132,992        | 2.75        | 161.5         | 0.9771        |
| <b>MoR (token-adaptive)</b>    | <b>132,992</b> | <b>2.69</b> | <b>159.1</b>  | <b>0.9885</b> |

*Notes.* Param ratio in Table 1 is independent/shared transformer parameters. “Transf. params” counts only the recurrent transformer block (the full model incl. embeddings/heads is  $\sim$ 712K for shared variants,  $\sim$ 1.11M for the independent stack). Compute-saving for the early-exit and MoR variants is  $\approx$ 40% of the fixed depth-4 FLOP budget. Token-reduction at the marker stage additionally compresses 4000 input genes to 256 retained markers (93.6% token reduction) at 98.5% accuracy.

## Dataset tasks and per-task results

The model is evaluated on the **unified TCGA benchmark**: 2,738 patient samples, 20,530 genes (top 4,000 highly-variable genes used as input tokens). The same recursive architecture (shared block, depth 4, 256 markers) is trained per task. Six prediction targets span tumor identity, clinical staging, and outcome.

**Table 4: Cohort-wise accuracy of the three model types, with parameters.** `cancer_type` is modeled *unified* (pooled across all four cohorts, it is the cohort identity); the five clinical / outcome targets are modeled *separately within each cohort* (breast, lung, thyroid, head&neck), for all three model types. Recurrent-block parameters are listed under each model. Best per row in **bold**.

| Cohort          | Task                          | Test accuracy by model      |                                    |                     |
|-----------------|-------------------------------|-----------------------------|------------------------------------|---------------------|
|                 |                               | Independent<br>530,432 par. | Recursive (shared)<br>132,992 par. | MoR<br>132,992 par. |
| Unified (all)   | <code>cancer_type</code>      | <b>0.9904</b>               | 0.9847                             | 0.9885              |
| Breast (867)    | <code>pathologic_stage</code> | <b>0.4195</b>               | 0.3506                             | 0.3333              |
|                 | <code>pathologic_T</code>     | <b>0.5632</b>               | 0.5287                             | 0.5287              |
|                 | <code>pathologic_N</code>     | 0.3448                      | <b>0.4540</b>                      | 0.3333              |
|                 | <code>os_binary</code>        | 0.6782                      | 0.5345                             | <b>0.7299</b>       |
|                 | <code>tumor_status</code>     | <b>0.8218</b>               | 0.7069                             | 0.7356              |
| Lung (943)      | <code>pathologic_stage</code> | 0.4180                      | <b>0.4603</b>                      | 0.4392              |
|                 | <code>pathologic_T</code>     | 0.4233                      | <b>0.4868</b>                      | 0.4709              |
|                 | <code>pathologic_N</code>     | 0.4603                      | <b>0.5556</b>                      | <b>0.5556</b>       |
|                 | <code>os_binary</code>        | 0.7302                      | 0.7778                             | <b>0.8519</b>       |
|                 | <code>tumor_status</code>     | 0.6032                      | 0.5079                             | <b>0.6296</b>       |
| Thyroid (505)   | <code>pathologic_stage</code> | 0.3168                      | 0.2376                             | <b>0.3564</b>       |
|                 | <code>pathologic_T</code>     | <b>0.3564</b>               | <b>0.3564</b>                      | 0.3267              |
|                 | <code>pathologic_N</code>     | <b>0.6040</b>               | 0.5446                             | 0.5743              |
|                 | <code>os_binary</code>        | <b>0.9505</b>               | <b>0.9505</b>                      | <b>0.9505</b>       |
|                 | <code>tumor_status</code>     | <b>0.7426</b>               | <b>0.7426</b>                      | 0.7030              |
| Head&neck (423) | <code>pathologic_stage</code> | 0.3059                      | 0.4824                             | <b>0.5765</b>       |
|                 | <code>pathologic_T</code>     | 0.2235                      | <b>0.3294</b>                      | 0.3176              |
|                 | <code>pathologic_N</code>     | 0.4588                      | <b>0.4824</b>                      | 0.3059              |
|                 | <code>os_binary</code>        | <b>0.9059</b>               | <b>0.9059</b>                      | <b>0.9059</b>       |
|                 | <code>tumor_status</code>     | <b>0.5882</b>               | 0.4706                             | 0.4588              |

*Notes.* Parameters under each header are recurrent transformer-block parameters (total ~1.11M Independent; ~712K shared and MoR). All numbers are single-seed (seed 42) test accuracies. `cancer_type` is the unified four-cohort model (`results/{independent,main,mor_token}.json`); the per-cohort clinical/outcome results are from `results-cohorts/{_independent, _mor_token}.json`. Cohort sizes (full, in parentheses) are split 65/15/20 train/val/test per cohort; with as few as 423 samples and some rare classes, per-cohort accuracies are noisier than the pooled numbers and no single model type dominates across cohorts.